

PHASE 1: COMPLETE STUDY GUIDE

Percentage + Profit & Loss + Average | Target: 1 Hour | Learn + Practice Together

TOPIC 1: PERCENTAGE (20 minutes)

1A. THE BASICS (2 min read)

What is a Percentage?

Percentage means 'per hundred'. It is a way to express a number as a fraction of 100.

$$X\% \text{ of } Y = (X / 100) \times Y$$

Worked Example (from your PDF Q1):

Find 20% of 730

$$= (20/100) \times 730 = 0.2 \times 730 = \mathbf{146}$$

Shortcut: 20% = 1/5. So 730/5 = 146. Fractions are faster!

Converting Between %, Fractions & Decimals

% to Fraction: Divide by 100 | Fraction to %: Multiply by 100 | Decimal to %: Multiply by 100

MEMORIZE THIS TABLE - it saves you 30+ seconds per question:

Percentage	Fraction	Decimal	Percentage	Fraction	Decimal
10%	1/10	0.1	25%	1/4	0.25
12.5%	1/8	0.125	33.33%	1/3	0.333
16.67%	1/6	0.167	50%	1/2	0.5
20%	1/5	0.2	75%	3/4	0.75
40%	2/5	0.4	80%	4/5	0.8
60%	3/5	0.6	66.67%	2/3	0.667

1B. PERCENTAGE INCREASE & DECREASE (3 min read)

The Multiplier Approach (FASTEST method)

Instead of calculating the percentage and adding/subtracting, just MULTIPLY:

X% increase => Multiply by $(1 + X/100)$

X% decrease => Multiply by $(1 - X/100)$

20% increase = $\times 1.2$ | 30% decrease = $\times 0.7$ | 15% increase = $\times 1.15$

Worked Example (from your PDF Q8):

Salary increased by 20%. New salary = Rs. 72,000. Find original salary.

$$\text{Original} \times 1.2 = 72,000$$

$$\text{Original} = 72,000 / 1.2 = \mathbf{Rs. 60,000}$$

Worked Example (from your PDF Q9):

Two numbers are 20% and 50% more than a third. First is what % of second?

Let third number = 100. First = 120, Second = 150.

$$\text{First as \% of Second} = (120/150) \times 100 = \mathbf{80\%}$$

Shortcut: Always let the base = 100. It eliminates fractions instantly.

1C. THE 'R% MORE OR LESS' FORMULA (3 min read)

This is the **MOST TESTED** concept in percentage. Learn these two by heart:

If A is R% MORE than B, then B is LESS than A by: $100 \times R / (100 + R) \%$

If A is R% LESS than B, then B is MORE than A by: $100 \times R / (100 - R) \%$

Worked Example (from your PDF Q10):

Ram's income is 20% less than Shyam's. How much % is Shyam's more than Ram's?

Apply formula: $100 \times 20 / (100 - 20) = 2000 / 80 = \mathbf{25\%}$

Worked Example (from your PDF Q11):

Price increased by 25%. By what % should consumption decrease to keep expenditure same?

Apply: $100 \times 25 / (100 + 25) = 2500 / 125 = \mathbf{20\%}$

Why this works: Expenditure = Price x Consumption. If price goes up by R%, consumption must go down by $100R/(100+R)\%$ for the product to stay same.

1D. SUCCESSIVE CHANGE (3 min read)

When a value changes by a% first, then by b%:

Net change = $a + b + (a \times b)/100 \%$

Use positive values for increase, negative for decrease.

Worked Example (from your PDF Q13):

A number is increased by 20%, then decreased by 10%. Net change?

Net = $20 + (-10) + (20 \times -10)/100 = 20 - 10 - 2 = \mathbf{+8\% \text{ increase}}$

SPECIAL CASE: Same % up then down (or vice versa):

Net change = $-R^2/100 \%$ (ALWAYS a decrease!)

Worked Example (from your PDF Q12):

A number decreased by 40%, then increased by 40%. Net change?

Net = $-(40)^2/100 = -1600/100 = \mathbf{-16\% \text{ (decrease)}}$

Trap Alert: Many students think +40% and -40% cancel out. They DON'T! There's always a net loss.

1E. PRICE, CONSUMPTION & EXPENDITURE (2 min read)

Core Relationship:

Expenditure = Price x Consumption

If price increases by R% and you want expenditure to increase by only X%:

Then consumption must change by: use successive change formula with price change = R and consumption change = unknown, set net = X

Worked Example (from your PDF Q15):

Price of apples increases by 30%. A man buys 6 kg less for Rs. 300. Find new price/kg.

30% of Rs. 300 = Rs. 90 = cost of 6 kg at new price.

New price per kg = $90/6 = \mathbf{Rs. 15/kg}$

Worked Example (from your PDF Q16):

Price increases by 20%. By what % reduce consumption so expenditure rises by only 8%?

Let consumption reduce by x%. Using successive change:

$20 + (-x) + 20(-x)/100 = 8$

$20 - x - 0.2x = 8 \Rightarrow 1.2x = 12 \Rightarrow x = \mathbf{10\%}$

1F. ELECTIONS & VOTES (1 min read)

Total Voters = Valid Votes + Invalid Votes + Non-Voters

Winner's Votes - Loser's Votes = Majority

Worked Example (from your PDF Q19):

Winner won by 24,000 votes. 10% didn't vote. Winner had 48% of total voters.

Let total voters = x. Winner = $0.48x$. Loser = $0.90x - 0.48x = 0.42x$.

Majority: $0.48x - 0.42x = 0.06x = 24,000 \Rightarrow x = \mathbf{4,00,000}$

PERCENTAGE: PRACTICE QUESTIONS

P1. Find 8% of 250.

- a) 18
- b) 20
- c) 22
- d) 25

P2. M is 80% of N. N is what percent of M?

- a) 80%
- b) 100%
- c) 120%
- d) 125%

P3. A number is increased by 20%, then decreased by 10%. Net % change?

- a) 10% increase
- b) 8% increase
- c) 12% increase
- d) 8% decrease

P4. In a 300 litre mixture there is 10% water. How much water to add to make it 25%?

- a) 50 litres
- b) 55 litres
- c) 60 litres
- d) 75 litres

P5. A gets 28% marks and fails by 18 marks. B gets 41% and has 21 marks more than pass marks. Find max marks.

- a) 200
- b) 250
- c) 300
- d) 350

P6. Shyam's salary is 25% more than Ram's. By what % is Ram's salary less than Shyam's?

- a) 25%
- b) 20%
- c) 15%
- d) 30%

PERCENTAGE: SOLUTIONS

P1. Answer: b) 20 | $8\% \text{ of } 250 = 8 \times 2.5 = 20$. Or: $250 \times \frac{8}{100} = 20$.

P2. Answer: d) 125% | Let $N = 100$, $M = 80$. N as % of $M = \frac{(100/80) \times 100}{1} = 125\%$.

P3. Answer: b) 8% increase | $\text{Net} = 20 + (-10) + \frac{(20 \times -10)}{100} = 20 - 10 - 2 = 8\% \text{ increase}$.

P4. Answer: c) 60 litres | Water = 30L. Let x added. $\frac{(30+x)}{(300+x)} = \frac{25}{100}$. $100(30+x) = 25(300+x)$. $3000+100x = 7500+25x$. $75x = 4500$. $x = 60\text{L}$.

P5. Answer: c) 300 | Pass marks = $28\% \text{ of } M + 18 = 41\% \text{ of } M - 21$. So $13\% \text{ of } M = 39$. $M = 300$.

P6. Answer: b) 20% | Apply formula: $100 \times \frac{25}{(100+25)} = \frac{2500}{125} = 20\%$.

TOPIC 2: PROFIT AND LOSS (20 minutes)

2A. THE THREE PRICES - CP, SP, MP (2 min read)

Understand these 3 prices first. Everything else builds on them:

CP (Cost Price) = What you PAID to buy the item (includes all expenses)

SP (Selling Price) = What you RECEIVED when you sold the item

MP (Marked Price) = Price WRITTEN ON THE TAG (before any discount)

The Chain:

CP --- (profit/loss) ---> SP --- (discount) --- MP

SP > CP => Profit | SP < CP => Loss | MP > SP => Discount was given

2B. THE 6 CORE FORMULAS (3 min - MEMORIZE ALL)

Profit = SP - CP | Loss = CP - SP

Profit% = (Profit / CP) x 100 | Loss% = (Loss / CP) x 100

Discount = MP - SP | Discount% = (Discount / MP) x 100

CRITICAL: Profit/Loss % is ALWAYS on CP. Discount % is ALWAYS on MP. Never mix these up!

The Multiplier Formulas (same concept as percentage!):

SP = CP x (100 + Profit%) / 100 | CP = SP x 100 / (100 + Profit%)

SP = MP x (100 - Discount%) / 100 | MP = SP x 100 / (100 - Discount%)

Worked Example (from your PDF Q1):

SP = 600, CP = 400. Find Profit%.

Profit = 600 - 400 = 200. Profit% = (200/400) x 100 = **50%**

Worked Example (from your PDF Q4):

SP = 54000, Profit = 12.5%, Discount = 10%. Find MP and CP.

SP = 90% of MP => 54000 = 0.9 x MP => **MP = Rs. 60,000**

SP = 112.5% of CP => 54000 = 1.125 x CP => **CP = Rs. 48,000**

Worked Example (from your PDF Q5):

CP = Rs.1700, wants 40% profit after 30% discount. Find MP.

SP = 1700 x 1.4 = Rs. 2380. SP = 70% of MP. MP = 2380/0.7 = **Rs. 3400**

2C. THE 'TWO ARTICLES' TRAP (3 min read)

These are favorite exam questions. There are TWO cases:

Case 1: Two articles sold at SAME SP, one at R% profit, other at R% loss

Result: ALWAYS A LOSS = $R^2/100$ %

Worked Example (from your PDF Q7):

Two articles sold at Rs. 32525 each. One at 30% gain, other at 30% loss.

Net loss = $30^2/100 = 900/100 =$ **9% loss**

Case 2: Two articles bought at SAME CP, one at R% profit, other at R% loss

Result: ALWAYS NO PROFIT NO LOSS

Worked Example (from your PDF Q8):

Two articles bought at Rs. 42250 each. One sold at 50% gain, other at 50% loss.

Profit on first = Loss on second. **Net = Zero**

Memory trick: Same SP = Loss (both start with S). Same CP = No change.

2D. SUCCESSIVE DISCOUNTS (2 min read)

Two successive discounts of a% and b%:

$$\text{Final SP} = \text{MP} \times (1 - a/100) \times (1 - b/100)$$

$$\text{Single equivalent discount} = a + b - ab/100 \%$$

Worked Example (from your PDF Q10):

Single discount equivalent to successive 10%, 30% and 20%?

$$\text{SP} = 0.9 \times 0.7 \times 0.8 \times \text{MP} = 0.504 \text{ MP}$$

$$\text{Single discount} = 1 - 0.504 = 0.496 = \mathbf{49.6\%}$$

Worked Example (from your PDF Q9 - comparing discounts):

Which is better for shopkeeper: (A) 20%, 30% or (B) 40%, 10%?

$$\text{A: } 0.8 \times 0.7 = 0.56 \Rightarrow 44\% \text{ discount} \mid \text{B: } 0.6 \times 0.9 = 0.54 \Rightarrow 46\% \text{ discount}$$

Option A gives less discount $\Rightarrow$ **A is better for shopkeeper**

2E. DISHONEST TRADER (2 min read)

A shopkeeper claims to sell at CP but cheats on weight:

$$\text{Gain\%} = (\text{True Weight} - \text{False Weight}) / \text{False Weight} \times 100$$

Worked Example (from your PDF Q12):

Sells at CP but uses 750 gm instead of 1 kg.

$$\text{Gain} = (1000 - 750)/750 \times 100 = 250/750 \times 100 = \mathbf{33.33\% \text{ profit}}$$

What if he also claims a loss?

Worked Example (from your PDF Q13):

Claims 8% loss but uses 800 gm for 1 kg.

He charges 92% of 1 kg price for 800 gm. Effective SP per 800 gm = Rs. 920 (if 1 kg = Rs. 1000).

$$\text{Actual CP of 800 gm} = \text{Rs. 800. Profit} = (920 - 800)/800 \times 100 = \mathbf{15\% \text{ profit}}$$

PROFIT & LOSS: PRACTICE QUESTIONS

PL1. SP = 600, CP = 400. Find Profit%.

- a) 33.33%
- b) 50%
- c) 25%
- d) 40%

PL2. By selling chocolates at Rs. 350, profit is 3/2 times the profit at Rs. 300. Find CP.

- a) Rs. 175
- b) Rs. 200
- c) Rs. 400
- d) Rs. 150

PL3. Two articles sold at Rs. 6000 each. One at 20% profit, other at 20% loss. Net result?

- a) No profit no loss
- b) 4% loss
- c) 4% profit
- d) 2% loss

PL4. Successive discounts of 20% and 10%. Equivalent single discount?

- a) 30%
- b) 28%
- c) 32%
- d) 25%

PL5. A man bought 2 articles for Rs. 1500. Sold one at 50% profit, other at 25% loss. Net = zero. Cost of costlier article?

- a) Rs. 500
- b) Rs. 1000
- c) Rs. 600

d) Rs. 900

PL6. Shopkeeper uses 800 gm weight instead of 1 kg and sells at CP. His profit?

- a) 20%
- b) 25%
- c) 30%
- d) 15%

PROFIT & LOSS: SOLUTIONS

PL1. Answer: b) 50% | Profit = 200. Profit% = $(200/400) \times 100 = 50\%$.

PL2. Answer: b) Rs. 200 | $350 - CP = 1.5(300 - CP)$. $350 - CP = 450 - 1.5CP$. $0.5CP = 100$. $CP = 200$.

PL3. Answer: b) 4% loss | Same SP case: Net loss = $R^2/100 = 400/100 = 4\%$ loss.

PL4. Answer: b) 28% | Single = $20 + 10 - (20 \times 10)/100 = 30 - 2 = 28\%$.

PL5. Answer: b) Rs. 1000 | Let CP of first = x. $0.5x = 0.25(1500 - x)$. $0.5x = 375 - 0.25x$. $0.75x = 375$. $x = 500$. Other = 1000.

PL6. Answer: b) 25% | Gain = $(1000 - 800)/800 \times 100 = 200/800 \times 100 = 25\%$.

TOPIC 3: AVERAGE (20 minutes)

3A. THE CORE FORMULA (2 min read)

Average = Sum of all observations / Number of observations

Sum = Average x Number (rearranged - use this form in most problems!)

Worked Example (from your PDF Q1):

Spent Rs. 10, 20, 30, 40, 60 over 5 days. Average?

Sum = 160. Average = $160/5 = \text{Rs. } 32$

Key insight (from your PDF Q2):

Average of 20 numbers is 0. At most how many can be > 0 ?

Answer: **19**. If 19 are positive, the 20th can be negative (equal to their sum) to make total = 0.

3B. THREE POWERFUL SHORTCUTS (3 min read)

Shortcut 1: Average of first n natural numbers

Average = $(n + 1) / 2$

Worked Example (from your PDF Q5):

Average of first 41 natural numbers = $(41+1)/2 = 42/2 = 21$

Shortcut 2: Average of AP (equal difference) series

Average = Middle term (if odd count) or Average of two middle terms (if even count)

Worked Example (from your PDF Q4):

Average of 27, 30, 33, 36, 39, 42, 45 = Middle term = **36**

No calculation needed! Just pick the middle number.

Shortcut 3: Assumed Mean Method

Pick a 'guess' average (A). Calculate deviations from A. Average = $A + (\text{sum of deviations})/n$

Worked Example (from your PDF Q3):

Marks: 76, 77, 79, 71, 72. Let assumed mean = 75.

Deviations: +1, +2, +4, -4, -3 = 0. Average = $75 + 0/5 = 75$

When to use: When numbers are close together and big. Much faster than adding all of them.

3C. WHEN THINGS CHANGE (3 min read)

Rule 1: If every number increases/decreases by k:

New Average = Old Average $\pm$ k

Worked Example (from your PDF Q7):

Average of 75 numbers is 35. Each increased by 5. New average = $35 + 5 = 40$

Rule 2: If every number multiplied by k:

New Average = k x Old Average

Worked Example (from your PDF Q8):

Average of 10 numbers = x. Each increased by 10%. New avg = $1.1x = x$ increased by 10%

3D. ERROR-BASED QUESTIONS (2 min read)

If a wrong value was used instead of the correct value:

Correct Average = Wrong Average + $(\text{Correct Value} - \text{Wrong Value}) / n$

Worked Example (from your PDF Q9):

Average of 40 students = 62.15. One student's marks read as 59 instead of 95.

Correct avg = $62.15 + (95 - 59)/40 = 62.15 + 36/40 = 62.15 + 0.9 = \mathbf{63.05}$

Don't recalculate the entire sum! Just add the correction.

3E. REPLACEMENT QUESTIONS (2 min read)

When one person is replaced and average changes:

New person's value = Old person's value + (Change in avg x n)

Worked Example (from your PDF Q10):

Average weight of 20 persons = 26 kg. A 14 kg person replaced. New avg = 27.25 kg.

Change in total = $(27.25 - 26) \times 20 = 1.25 \times 20 = 25$.

New person = $14 + 25 = \mathbf{39 \text{ kg}}$

3F. WEIGHTED AVERAGE (3 min read)

When different groups have different sizes:

Weighted Avg = $(n_1 \times \text{avg}_1 + n_2 \times \text{avg}_2 + \dots) / (n_1 + n_2 + \dots)$

Worked Example (from your PDF Q11):

4 classes: 10, 20, 30, 40 students. Pass rates: 20%, 30%, 60%, 100%.

Pass% = $(10 \times 20 + 20 \times 30 + 30 \times 60 + 40 \times 100) / (10 + 20 + 30 + 40)$

= $(200 + 600 + 1800 + 4000) / 100 = 6600/100 = \mathbf{66\%}$

Worked Example (from your PDF Q12):

16 boys avg 51 kg, 8 boys avg 36 kg. Overall avg?

= $(16 \times 51 + 8 \times 36) / 24 = (816 + 288) / 24 = 1104/24 = \mathbf{46 \text{ kg}}$

Worked Example (from your PDF Q14 - finding unknown group):

30 officers avg salary Rs.120, labourers avg Rs.40, firm avg Rs.50. Total employees?

Let labourers = x. $(30 \times 120 + 40x) / (30 + x) = 50$

$3600 + 40x = 1500 + 50x \Rightarrow 10x = 2100 \Rightarrow x = 210$. Total = **240**

AVERAGE: PRACTICE QUESTIONS

A1. Average of first 50 natural numbers?

- a) 25
- b) 25.5
- c) 26
- d) 26.5

A2. Average of 15, 18, 21, 24, 27, 30, 33?

- a) 21
- b) 24
- c) 27
- d) 30

A3. Average of 30 students was 72. One mark was 85 instead of 55. Correct average?

- a) 71
- b) 70
- c) 73
- d) 72.5

A4. Average weight of 16 boys = 50.25 kg, 8 boys = 45.15 kg. Overall?

- a) 47.55 kg
- b) 48.55 kg
- c) 49.25 kg
- d) 46.70 kg

A5. 7 consecutive integers. Average of first 5 is n. Average of all 7?

- a) n

- b) $n+1$
- c) $n-1$
- d) $(n+2)/7$

A6. Avg temperature Mon-Wed = 36.5. Avg Tue-Thu = 38.5. Monday temp = $6/7$ of Thursday. Thursday temp?

- a) 35
- b) 42
- c) 38
- d) 40

AVERAGE: SOLUTIONS

A1. Answer: b) 25.5 | Avg of first n natural numbers = $(n+1)/2 = 51/2 = 25.5$.

A2. Answer: b) 24 | AP with 7 terms. Middle term = 4th term = 24.

A3. Answer: a) 71 | Correct avg = $72 + (55-85)/30 = 72 - 1 = 71$.

A4. Answer: b) 48.55 kg | $= (16 \times 50.25 + 8 \times 45.15)/24 = (804 + 361.2)/24 = 48.55$.

A5. Answer: b) $n+1$ | First 5: avg = 3rd term = n . All 7: avg = 4th term = $n+1$ (consecutive, so $+1$).

A6. Answer: b) 42 | Sum MTW = 109.5. Sum TWTh = 115.5. Thu - Mon = 6. Mon = $6/7$ Thu. Thu - $6/7$ Thu = 6. Thu/ 7 = 6. Thu = 42.

PHASE 1: QUICK REVISION CARD

PERCENTAGE - 5 Formulas to Remember

1. X% of Y = $XY/100$
2. R% more: other is less by $100R/(100+R)\%$
3. R% less: other is more by $100R/(100-R)\%$
4. Successive change: $a + b + ab/100$
5. Same % up then down: Net = $-R^2/100\%$ (always loss)

PROFIT & LOSS - 5 Formulas to Remember

1. Profit% = $(\text{Profit}/\text{CP}) \times 100$ | Discount% = $(\text{Discount}/\text{MP}) \times 100$
2. $\text{SP} = \text{CP}(100+P\%)/100$ | $\text{SP} = \text{MP}(100-D\%)/100$
3. Same SP, R% profit & R% loss: Net loss = $R^2/100\%$
4. Same CP, R% profit & R% loss: No profit no loss
5. Dishonest trader: Gain = $(\text{True}-\text{False})/\text{False} \times 100\%$

AVERAGE - 5 Formulas to Remember

1. Average = Sum / n | $\text{Sum} = \text{Avg} \times n$
2. First n natural numbers: Avg = $(n+1)/2$
3. AP series: Avg = middle term
4. Error fix: New avg = $\text{Old avg} + (\text{correct} - \text{wrong})/n$
5. Weighted avg = $(n1.a1 + n2.a2) / (n1 + n2)$

Done with Phase 1? Move to Phase 2: Numbers, Permutation & Combination, Probability